Academic Timetable Generator

Project Report – CSP Backtracking Approach

1. Role of Files

File / Folder	Role
app.py	Flask backend. Serves the web page, receives input, runs the CSP backtracking solver and returns the timetable (also generates CSV download).
templates/	Folder required by Flask to store HTML pages.
templates/index.html	Frontend user interface: input form, subject/teacher tables, and timetable display (class and teacher views).

2. How to Run the Program

1. Install Python 3.8 or above.
2. Open a terminal in the project folder (the folder containing app.py).
3. Install Flask:

```
pip install flask
```

4. Run the program:

```
python app.py
```

5. Open the browser and go to http://127.0.0.1:5000
6. Enter / load the input data and click “Generate Timetable”.

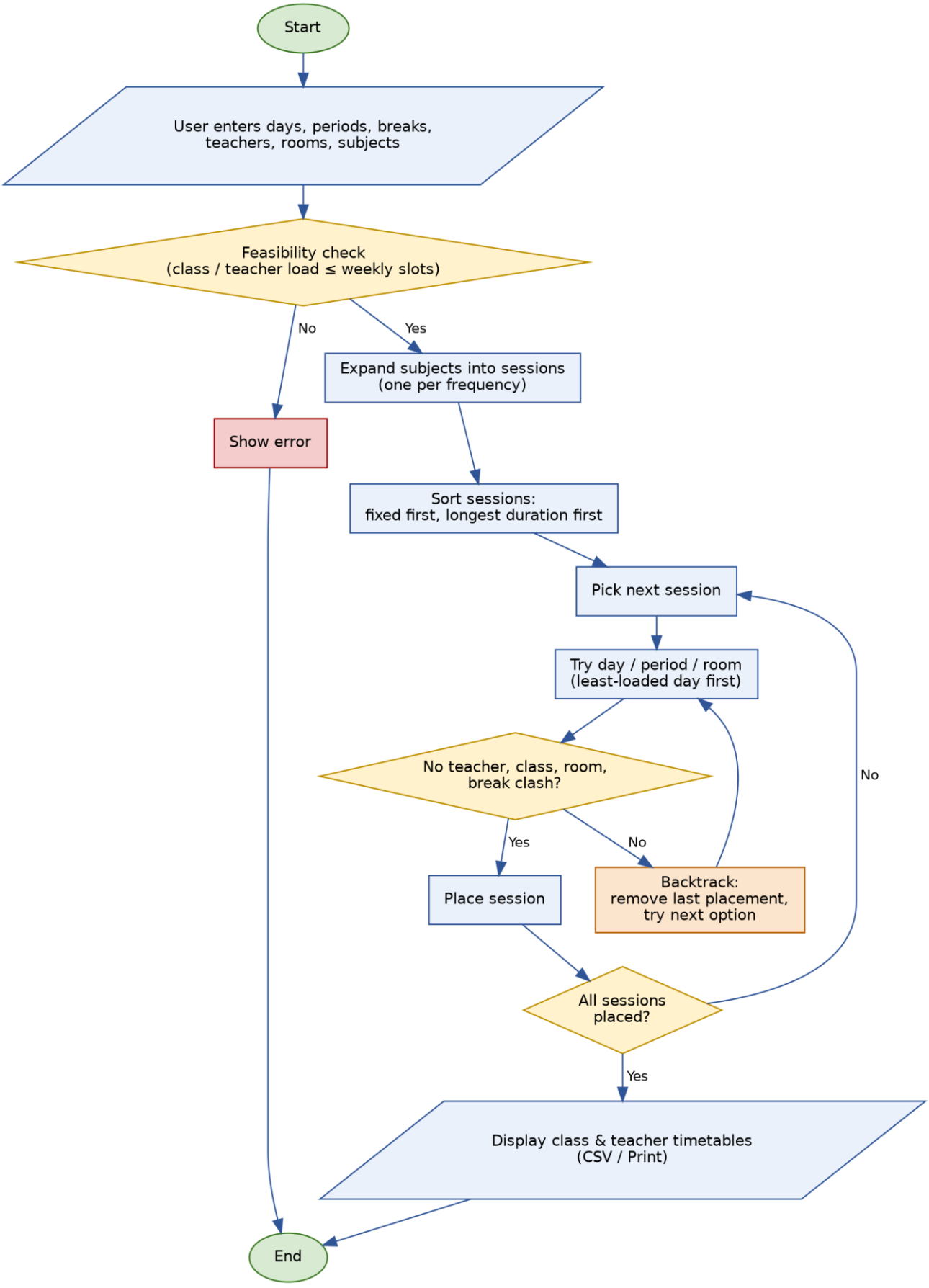
3. Algorithm

The timetable is solved as a Constraint Satisfaction Problem using recursive backtracking.

1. Check feasibility (class and teacher load must fit within weekly slots).
2. Expand each subject into sessions (one session per frequency).
3. Sort sessions: fixed-slot sessions first, then longest duration first.
4. For each session, try day / period / room (least-loaded day first).
5. Place it only if there is no teacher, class, room or break clash.
6. If no valid slot exists, backtrack: undo the previous placement and try the next option.
7. When all sessions are placed, output class and teacher timetables.

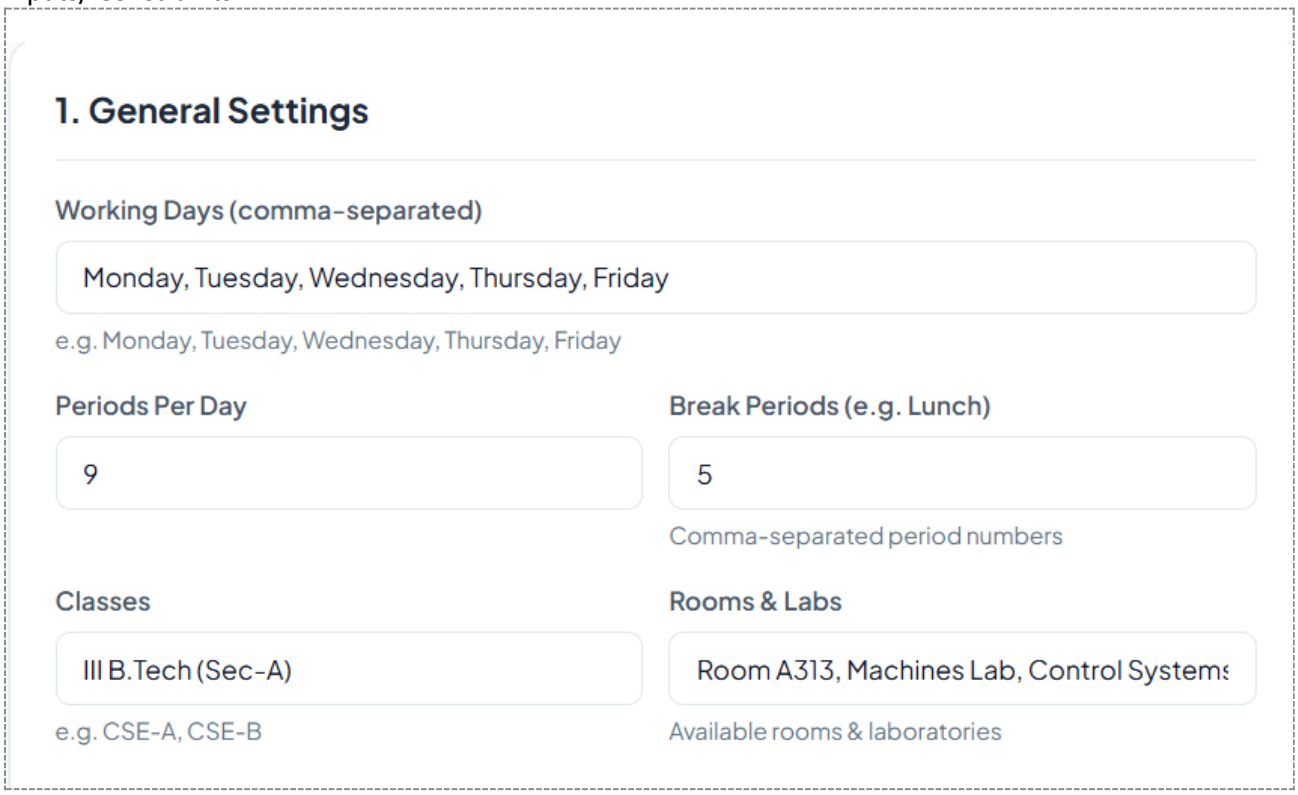
Constraints handled: no teacher / class / room clash, no session in break periods, teacher unavailability, multi-period lab blocks, and parallel sub-batch labs.

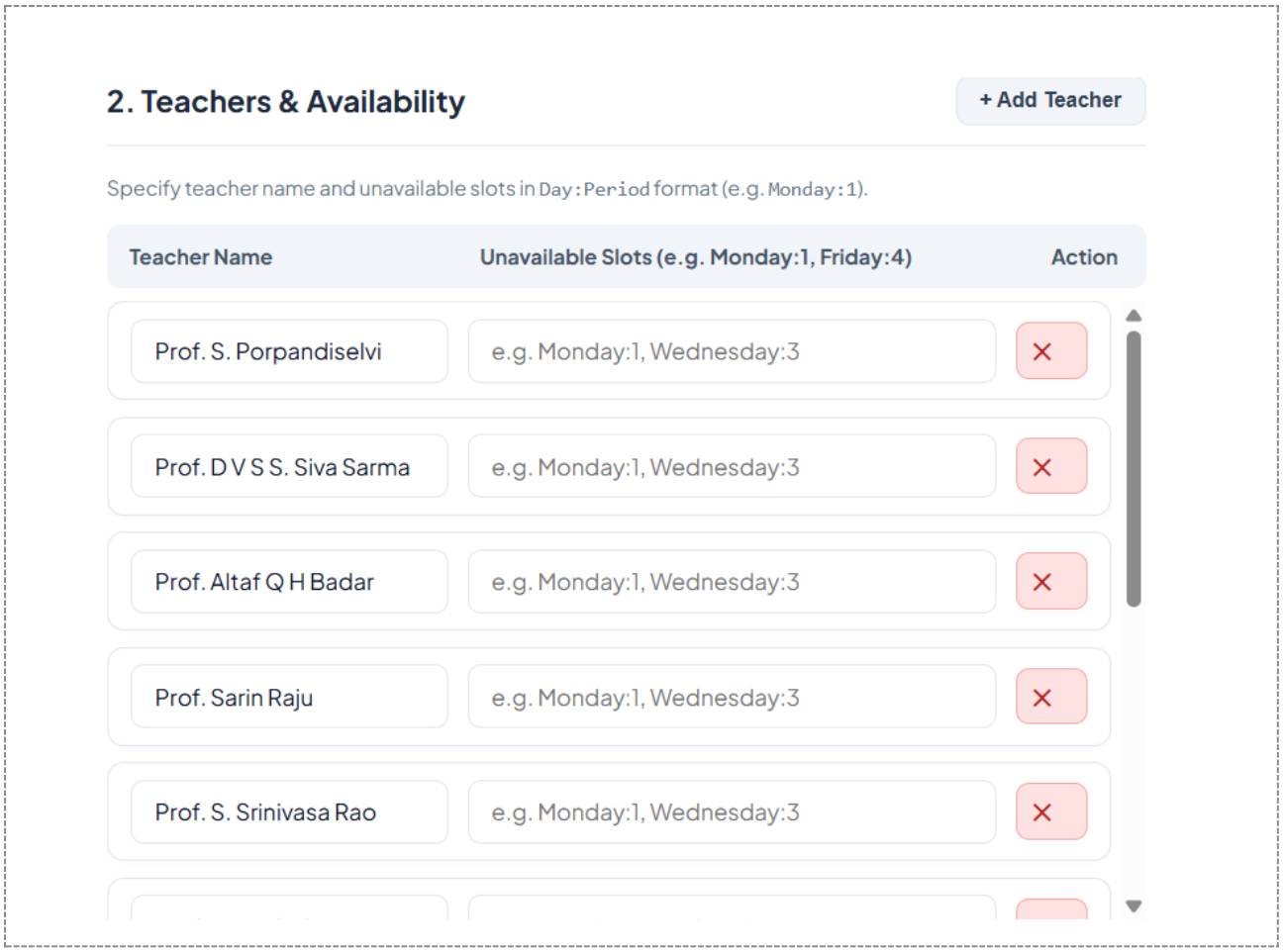
4. Flowchart


Start
User enters days, periods, breaks, teachers, rooms, subjects
Feasibility check (class / teacher load ≤ weekly slots)
No
Yes
Expand subjects into sessions (one per frequency)
Show error
Sort sessions: fixed first, longest duration first
Pick next session
Try day / period / room (least-loaded day first)
No teacher, class, room, break clash?
Yes
No
Place session
Backtrack: remove last placement, try next option
No
All sessions placed?
Yes
Display class & teacher timetables (CSV / Print)
End

5. Screenshots

Inputs/ Constraints





2. Teachers & Availability

+ Add Teacher

Specify teacher name and unavailable slots in `Day:Period` format (e.g. `Monday:1`).

Teacher Name	Unavailable Slots (e.g. Monday:1, Friday:4)	Action
Prof. A. Kirubakaran	e.g. Monday:1, Wednesday:3	✕
Prof. P. Srinivasan	e.g. Monday:1, Wednesday:3	✕
Prof. N. Vishwanathan	e.g. Monday:1, Wednesday:3	✕
Prof. G. Siva Kumar	e.g. Monday:1, Wednesday:3	✕
Prof. A.V. Giridhar	e.g. Monday:1, Wednesday:3	✕
Prof. Palash Mishra	e.g. Monday:1, Wednesday:3	✕

3. Subjects & Requirements

+ Add Subject

Define each subject, class, assigned teacher, weekly frequency, duration (consecutive periods e.g. Labs = 2, Workshop = 3), room, and optional fixed slot (e.g. `Friday:5`).

Subject Name	Class	Assigned Teacher	Freq / Wk	Duration (Periods)	Room / Lab (Optional)	Fixed Slot (Optional)	Action
EE1301 Power Electronics (Slc	III B.Tech (Sec-A	Prof. S. Porpandiselvi	3	1	Room A313	e.g. Friday:5	✕
EE1303 Microprocessors (Slo	III B.Tech (Sec-A	Prof. D V S S. Siva Sarma	3	1	Room A313	e.g. Friday:5	✕
EE1305 PSOC (Slot F)	III B.Tech (Sec-A	Prof. Altaf Q H Badar	3	1	Room A313	e.g. Friday:5	✕
MS1369 Management (Slot D)	III B.Tech (Sec-A	Prof. Sarin Raju	3	1	Room A313	e.g. Friday:5	✕
EE1321(E) Energy (Slot G)	III B.Tech (Sec-A	Prof. S. Srinivasa Rao	3	1	Room A313	e.g. Friday:5	✕
EE1307 Machines Lab (A1)	III B.Tech (Sec-A	Prof. A. Kirubakaran	1	3	Machines Lab	Monday:2	✕

RESET TO: 1. Default | 2. NITW (Fixed, No Minor) | 3. NITW (Auto Freq=3)

✨ Generate Timetable

Student TimeTable

Class: III B.Tech (Sec-A)	Teacher: Prof. A. Kirubakaran	Teacher: Prof. A.V. Giridhar	Teacher: Prof. Altaf Q H Badar	Teacher: Prof. D V S S. Siva Sarma	Teacher: Prof. G. Siva Kumar	Teacher: Prof. N. Vishwanathan	Teacher: Prof. P. Srinivasan	Teacher: Prof. Palash Mishra	Teacher: Prof. S. Porpandiselvi	Teacher: Prof. S. Srinivasa Rao

Day / Period	Period 1	Period 2	Period 3	Period 4	Period 5 (Break)	Period 6	Period 7	Period 8	Pe
Monday	EE1301 Power Electronics (Slot A) Prof. S. Porpandiselvi • Room A313	PARALLEL SUB-BATCHES (A1 & A2) EE1307 Machines Lab (A1) Prof. A. Kirubakaran • Machines Lab EE1309 Control Lab (A2) Prof. G. Siva Kumar • Control Systems Lab	PARALLEL SUB-BATCHES (A1 & A2) EE1307 Machines Lab (A1) Prof. A. Kirubakaran • Machines Lab EE1309 Control Lab (A2) Prof. G. Siva Kumar • Control Systems Lab	PARALLEL SUB-BATCHES (A1 & A2) EE1307 Machines Lab (A1) Prof. A. Kirubakaran • Machines Lab EE1309 Control Lab (A2) Prof. G. Siva Kumar • Control Systems Lab	BREAK	EE1305 PSOC (Slot F) Prof. Altaf Q H Badar • Room A313	EE1321(E) Energy (Slot G) Prof. S. Srinivasa Rao • Room A313	— Free —	— F
Tuesday	EE1309 Control Lab (A1) Prof. N. Vishwanathan • Control Systems Lab	EE1309 Control Lab (A1) Prof. N. Vishwanathan • Control Systems Lab	EE1309 Control Lab (A1) Prof. N. Vishwanathan • Control Systems Lab	EE1303 Microprocessors (Slot E) Prof. D V S S. Siva Sarma • Room A313	BREAK	EE1305 PSOC (Slot F) Prof. Altaf Q H Badar • Room A313	EE1321(E) Energy (Slot G) Prof. S. Srinivasa Rao • Room A313	— Free —	— F
Wednesday	EE1307 Machines Lab (A2) Prof. P. Srinivasan • Machines Lab	EE1307 Machines Lab (A2) Prof. P. Srinivasan • Machines Lab	EE1307 Machines Lab (A2) Prof. P. Srinivasan • Machines Lab	EE1303 Microprocessors (Slot E) Prof. D V S S. Siva Sarma • Room A313	BREAK	MS1369 Management (Slot D) Prof. Sarin Raju • Room A313	EE1321(E) Energy (Slot G) Prof. S. Srinivasa Rao • Room A313	— Free —	— F
Thursday	EE1303 MP Lab (A1) Prof. A.V. Giridhar • Microprocessors Lab	EE1303 MP Lab (A1) Prof. A.V. Giridhar • Microprocessors Lab	EE1301 Power Electronics (Slot A) Prof. S. Porpandiselvi • Room A313	EE1303 Microprocessors (Slot E) Prof. D V S S. Siva Sarma • Room A313	BREAK	MS1369 Management (Slot D) Prof. Sarin Raju • Room A313	EE1381 Fractal Course (Slot X) Prof. Palash Mishra • Room A313	— Free —	— F
Friday	EE1303 MP Lab (A2) Prof. A.V. Giridhar • Microprocessors Lab	EE1303 MP Lab (A2) Prof. A.V. Giridhar • Microprocessors Lab	EE1301 Power Electronics (Slot A) Prof. S. Porpandiselvi • Room A313	EE1305 PSOC (Slot F) Prof. Altaf Q H Badar • Room A313	BREAK	MS1369 Management (Slot D) Prof. Sarin Raju • Room A313	— Free —	— Free —	— F

Teachers Timetable

Class: III B.Tech (Sec-A) | Teacher: Prof. A. Kirubakaran | Teacher: Prof. A.V. Giridhar | **Teacher: Prof. Altaf Q H Badar** | Teacher: Prof. D V S S. Siva Sarma | Teacher: Prof. G. Siva Kumar | Teacher: Prof. N. Vishwanathan | Teacher: Prof. P. Srinivasan | Teacher: Prof. Palash Mishra | Teacher: Prof. S. Porpandiselvi | Teacher: Prof. S. Srinivasa Rao

Day / Period	Period 1	Period 2	Period 3	Period 4	Period 5 (Break)	Period 6	Period 7	Period 8	Period 9
Monday	— Free —	— Free —	— Free —	— Free —	BREAK	**EE1305 PSOC (Slot F)** Class: III B.Tech (Sec-A) • Room A313	— Free —	— Free —	— Free —
Tuesday	— Free —	— Free —	— Free —	— Free —	BREAK	**EE1305 PSOC (Slot F)** Class: III B.Tech (Sec-A) • Room A313	— Free —	— Free —	— Free —
Wednesday	— Free —	— Free —	— Free —	— Free —	BREAK	— Free —	— Free —	— Free —	— Free —
Thursday	— Free —	— Free —	— Free —	— Free —	BREAK	— Free —	— Free —	— Free —	— Free —
Friday	— Free —	— Free —	— Free —	**EE1305 PSOC (Slot F)** Class: III B.Tech (Sec-A) • Room A313	BREAK	— Free —	— Free —	— Free —	— Free —

Day / Period	Period 1	Period 2	Period 3	Period 4	Period 5 (Break)	Period 6	Period 7	Period 8	Peric
Monday	— Free —	— Free —	— Free —	— Free —	BREAK	— Free —	— Free —	— Free —	— Fre
Tuesday	— Free —	— Free —	— Free —	**EE1303 Microprocessors (Slot E)** Class: III B.Tech (Sec-A) • Room A313	BREAK	— Free —	— Free —	— Free —	— Fre
Wednesday	— Free —	— Free —	— Free —	**EE1303 Microprocessors (Slot E)** Class: III B.Tech (Sec-A) • Room A313	BREAK	— Free —	— Free —	— Free —	— Fre
Thursday	— Free —	— Free —	— Free —	**EE1303 Microprocessors (Slot E)** Class: III B.Tech (Sec-A) • Room A313	BREAK	— Free —	— Free —	— Free —	— Fre
Friday	— Free —	— Free —	— Free —	— Free —	BREAK	— Free —	— Free —	— Free —	— Fre